Exercises

Exercise 10.1 · Vocabulary. A graph G has vertices 1, 2, 3, 4, 5, 6 and edges 1-2, 1-3, 2-3, 3-4, 4-5, 4-6, 5-6.

1. Give the order of G and the degree of each vertex, and check the property of the sum of the degrees.
2. Is G complete? connected?
3. Give a walk of length 3 from 1 to 6 and a cycle through 5.

Exercise 10.2 · Handshakes.

1. Can one build a graph of order 7 all of whose vertices have degree 3?
2. Can six people each shake hands with exactly three others? If so, how many handshakes are there?

Exercise 10.3 · A directed graph. A directed graph has vertices A, B, C and arcs $A \to B$, $A \to C$, $B \to C$ and $C \to A$.

1. Write its adjacency matrix M (order A, B, C). Is it symmetric?
2. Compute M^2 and M^3.
3. How many paths of length 3 are there from A to A? Write them out.

Exercise 10.4 · A tournament. Eight teams play a tournament: each meets each of the others once. Model with a graph and determine the number of matches.

Exercise 10.5 · Degrees and the square of the matrix. A graph has adjacency matrix

$$M = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

1. Draw it.
2. Compute M^2 and compare its diagonal with the degrees of the vertices.
3. Prove that for every undirected graph, $\left(M^2\right)_{ii}$ is the degree of vertex i.

Exercise 10.6 · Testing connectedness. A graph has vertices 1 to 5 and edges 1-2, 2-3, 3-1, 4-5. Let M be its adjacency matrix.

1. Compute, with a calculator, $S = M + M^2 + M^3 + M^4$.
2. Is the graph connected? Justify directly, then check that this result can be read from the entries of S.

Exercise 10.7 · **Counting triangles.** A triangle of a graph is a set of three pairwise adjacent vertices.

1. Prove that the number of triangles is $\frac{1}{6}$ of the sum of the diagonal entries of M^3.
2. Apply this to the complete graph of order 4 and to the graph of exercise 10.5.